PnS.

Week-1:

* **Sample Mean:**

$$\bar{x} = \sum_{i=1}^{n} \frac{x_i}{n}$$

* **Sample Variance:**

$$S^2 = \sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{(n-1)}$$

* **Sample Standard deviation:**

$$S = \sqrt{\sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{(n-1)}}$$

* **Chebyshev's Inequality:**

$$P(|X - E(X)| \geq k) \leq \frac{Var(X)}{k^2}$$

$E(X) \to$ mean
$Var(X) \to$ Variance
$k \to$ +ve real number

$$\frac{|S_k|}{n} \geq 1 - \frac{n-1}{nk^2} > 1 - \frac{1}{k^2}$$

$S_k \to$ no. of elements in set S_k.

* **One-sided Chebyshev Inequality:**

$$\frac{N(k)}{n} \leq \frac{1}{1+k^2}$$

Week-2

Probability

① Sample Space: The set of all possible outcomes of an experiment.

② Events: Any subset E of the sample space.

Properties:

$E \cup F$ → Union → consists of all outcomes of E or F or both.

$E \cap F$ → Intersection → consists of all outcomes in both E and F.

ϕ → null event → event consisting of no outcomes.

$EF = \phi$ → mutually ~~exhaustive~~ exclusive.
→ implies that E and F cannot both occur.

E^c → E complement → consists of all outcomes in sample space S that are not in E.

$E \subset F \ (F \supset E)$ → all outcomes in E are also in F.

Rules:

① Commutative law:-
$E \cup F = F \cup E$ & $E \cap F = F \cap E$

② Associative law:- $(E \cup F) \cup G = E \cup (F \cup G)$ &
$(E \cap F) \cap G = E \cap (F \cap G)$

③ Distributive law:- $(E \cup F) \cap G = (E \cap G) \cup (F \cap G)$ &
$E \cap F \cup G = (E \cup G) \cap (F \cup G)$

④ De Morgan's law:- $(E \cup F)^c = E^c \cap F^c$ &
$(E \cap F)^c = E^c \cup F^c$

* **Axioms of Probability** :-

① $0 \leq P(E) \leq 1$

② $P(S) = 1$

③ For any sequence of mutually exclusive events $E_1, E_2 \ldots$

$$P\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{i=1}^{n} P(E_i), \; n = 1, 2, \ldots, \infty$$

Proposition - 1 :- $P(E^c) = 1 - P(E)$

Proposition - 2 :- $P(E \cup F) = P(E) + P(F) - P(E \cap F)$

* **Conditional probability** :- The condn. prob. of E given that F has occured is given by.

$$P(E|F) = \frac{P(E \cap F)}{P(F)} \quad [P(F) > 0]$$

* **Baye's Formula** :-

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

$P(A|B)$ = P(A) given B is true

$P(B|A)$ = P(B) given A is true.

Week-3

* Random Variables: Two types :- ① Discrete ② Continuous

Discrete R.V. :- Countably finite / infinite (0, 1, 2, ...)
Operations carried with Summation (Σ)

Continuous R.V. :- Measurable intervals or union of disjoint intervals (5.5, 5.7, ...)
operations carried with Integration ($\int$)

* Discrete Random Variable

Expectation (Mean) :-

$$E(X) = \sum_i x_i \, P\{X = x_i\}$$

Variance

$$Var(X) = E[(X-\mu)^2] \text{ or } Var(X) = E[X^2] - E[X]^2$$

Standard Variation

$$Std.dev.(X) = \sqrt{Var(X)}$$

$$Var(aX+b) = a^2 \, Var(X)$$
$$Var(aX) = a^2 \, Var(X)$$

Cumulative distribution function

$$F(X) = P\{X \le x\}$$

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* Continuous Random Variable:-

Probability Density Function:-

$$\int_{-\infty}^{\infty} f(x)\, dx = 1$$

Mean:-

$$E[X] = \int_{-\infty}^{\infty} x f(x)\, dx$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f(x)\, dx$$

Variance:-

$$Var[X] = E[X^2] - E[X]^2$$

week-4

* Binomial Distribution:-

Probability mass function:-

$$P\{X = i\} = \binom{n}{i} p^i (1-p)^{n-i}$$

$$\binom{n}{i} = \frac{n!}{i!\,(n-i)!}$$

Mean:-

$$E[X] = np$$

Variance:-

$$var(X) = np(1-p)$$

$n \rightarrow$ no. of independent trials
$p \rightarrow$ success probability
$(p-1) \rightarrow$ failure probability.

* Poisson's Distribution:-
when n is very very large and p is small.

$\lambda = np$

$B.D \cong P.D$

$$P[X=x] = \frac{e^{-\lambda}\cdot\lambda^{x}}{x!},\ x=0,1,2,\ldots$$

Mean:- $E(X) = \lambda$

Variance:- $Var(X) = \lambda$

week-5

* Normal Distribution:-

Mean:- $E[X] = \mu$

Variance:- $Var(X) = \sigma^2$

$\phi(-x) = P\{Z<-x\} = P\{Z>x\} = 1-\phi(x)$

$Z = \frac{X-\mu}{\sigma}$

$P(X<b) = P\left\{\frac{X-\mu}{\sigma} < \frac{b-\mu}{\sigma}\right\} = \phi\left(\frac{b-\mu}{\sigma}\right)$

* Uniform Distribution:-

$$P[a<X<b] = \frac{1}{\beta-\alpha}\int_a^b dx = \frac{b-a}{\beta-\alpha}$$

$E[X] = \frac{\alpha+\beta}{2}$, $Var(X) = \frac{(\beta-\alpha)^2}{12}$

* Gamma Dist^n:-

$E(X) = \frac{\alpha}{\lambda}$

$Var(X) = \frac{\alpha}{\lambda^2}$

* Expo^n Dist^n:-

$E[X] = 1/\lambda$

$Var(X) = \frac{1}{\lambda^2}$

* Beta Dist^n:-

$E[X] = \frac{\alpha}{\alpha+\beta}$

$Var(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$

Week-6:

* Sampling Distributions:

Mean: $E[\bar{x}] = \mu$.

Variance: $V(\bar{x}) = \frac{\sigma^2}{n}$

Std. deviation / Std. error:

$$\bar{x} = \frac{\sigma}{\sqrt{n}}$$

* Central limit theorem:

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$\bar{x} \rightarrow$ sample mean
$\mu \Rightarrow$ population mean
$n \rightarrow$ sample size
$\sigma \rightarrow$ std. dev.

* Central limit theorem is better applicable if sample size $n \geq 30$

* t-distribution:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

degree of freedom $\nu = n-1$
$s \rightarrow$ std. dev of sample

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* Chi-Square distⁿ :-

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

d.o.f ; $v = n-1$

$s^2 \rightarrow$ Variance of sample
$\sigma^2 \rightarrow$ variance of popⁿ.
$n \rightarrow$ sample size.

* F-distributionⁿ

$$F = \frac{S_1^2}{S_2^2}$$

d.o.f ; $v_1 = n_1 - 1$, $v_2 = n_2 - 1$

S_1^2 & $S_2^2 \rightarrow$ variance of independent random samples.

* Estimationⁿ

Sample mean, $\bar{x}$
population mean, μ
Sample proportion, p
population proportion, $P = \frac{x}{n}$

sample variance, S^2
population variance, $S_x^2 = \frac{n}{n-1} s^2$

Estimate of std. error $= \frac{S}{\sqrt{n}}$

Std. Normal Distribution for large n :-

$$P\left(-Z_{\alpha/2} \le \frac{\bar{X}-\mu}{\sigma/n} \le Z_{\alpha/2}\right) = 1-\alpha$$

Maxᵐ. error of estimateⁿ

$$E = Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

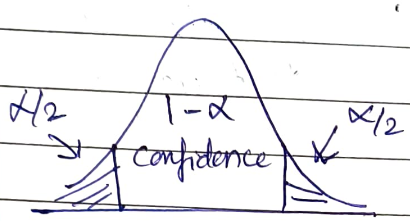
Maxᵐ. error of estimate, normal popⁿ., (σ unknown) :-

$$E = t_{\alpha/2} \cdot \frac{S}{\sqrt{n}}$$

Sample size determination

$$n = \left[\frac{Z_{\alpha/2} \cdot \sigma}{E}\right]^2$$

Interval Estimate



α : Shaded area

Interval Estimation : Mean

Large sample confidence Interval - μ (σ known)

$$\bar{x} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

large sample confidence Interval - μ

$$\bar{x} - Z_{\alpha/2} \frac{s}{\sqrt{n}} < \mu < \bar{x} + Z_{\alpha/2} \frac{s}{\sqrt{n}}$$

Small sample confidence interval - μ

$$\bar{x} - t_{\alpha/2} \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2} \frac{s}{\sqrt{n}}$$

Interval Estimation - Variance -

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2}} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}}$$

* **Interval Estimation : Proportion -**

$$\hat{p} - Z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} < p < \hat{p} + Z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

where,
$\hat{p} \rightarrow$ sample proportion.
$n \rightarrow$ sample size
$p \rightarrow$ population proportion.

$$\hat{p} = \frac{x}{n}$$

Week-7 -

* **Level of significance - (α)** [$\alpha = 0.10, 0.05, 0.01$]

Type-1 error - If null hyp. is rejected, when it is true

Type-2 error - If null hyp. is not rejected, when it is false.

Level of significance - Probability of committing a type-1 error

$P(\text{type-2 error}) = \beta$.

Actual sitn.	Decision: Accept H_0	Reject H_0
H_0 true	Correct $(1-\alpha)$	Error type 1 (α)
H_0 False	Error type 2 (β)	Correct $(1-\beta)$

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* Testing of Hyp :- One Mean (σ known) [Large Sample]

Test statistic value :

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \quad \text{or} \quad Z = \frac{\bar{x} - \mu}{S/\sqrt{n}}$$

(σ unknown)

$\bar{x}$ - Sample mean
μ - popn. mean
S - Std. deviation
σ^2 - Variance

* Testing of Hyp :- One mean - Small sample [σ unknown]

Test statistic value :

$$t = \frac{\bar{x} - \mu_0}{S/\sqrt{n}} \quad \text{where, } \nu = n-1$$

* Testing of Hyp :- Two means - large sample [Variance known]

Test statistic value :-

$$Z = \frac{\bar{x} - \bar{y} - \delta_0}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}}$$

Two means - large sample
[unknown popn. variance].

$$Z = \frac{\bar{X} - \bar{Y} - \delta_0}{\sqrt{\frac{S_x^2}{n} + \frac{S_y^2}{m}}}$$

* Testing of Hyp.: Two means - Small sample :-
[σ unknown]

$$S_p^2 = \frac{(n-1)S_x^2 + (m-1)S_y^2}{n+m-2}$$

Test statistics value n

$$t = \frac{\bar{X} - \bar{Y} - \delta_0}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}}$$

degree of freedom = $n+m-2$

Week-8:-

* Testing of Hyp. :- One proportion - large Sample

Null Hypotheses : $H_0 - p = p_0$

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \Rightarrow \hat{p} = \frac{x}{n}$$

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* Testing of Hypothesis :- Two proportions - large sample

Null Hyp. : $H_0 \Rightarrow P_1 - P_2 = 0$

$$Z = \frac{\hat{P}_1 - \hat{P}_2}{\sqrt{\hat{p}\,\hat{q}\left(\frac{1}{m} + \frac{1}{n}\right)}}$$

$$\hat{P}_1 = \frac{x_1}{n_1}, \quad \hat{P}_2 = \frac{x_2}{n_2}, \quad \hat{P} = \frac{x_1 + x_2}{n_1 + n_2}, \quad \hat{q} = 1 - \hat{P}$$

* Testing of Hyp :- Several proportions

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$O \rightarrow$ observed freq.

$E \rightarrow$ Expected freq.

Degree of freedom $= (r-1) \times (c-1)$

$r \rightarrow$ no. of rows

$c \rightarrow$ no. of columns

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Week-9:

* Covariance:

$$\text{Covariance}(X, Y) = \sum_{i=1}^{n} \frac{(X_i - \bar{X})(Y_i - \bar{Y})}{n-1}$$

Co-efficient of correlation (r):

$$r = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}}$$

$$= \frac{\sum_{i=1}^{n} (X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum_{i=1}^{n} (X - \bar{X})^2}\sqrt{\sum_{i=1}^{n} (Y - \bar{Y})^2}}$$

* Interpretation based on r value:

$r = -1$: perfect -ve corr^n.

$r = -0.99$ to -0.76 : Strong -ve corr^n.

$r = -0.75$ to -0.26 : Moderate -ve corr^n.

$r = -0.25$ to 0 : weak -ve corr^n.

$r = 0$: Zero corr^n

$r = 0$ to 0.25 : weak +ve corr^n.

$r = 0.26$ to 0.75 : Moderate +ve corr^n.

$r = 0.76$ to 0.99 : Strong +ve corr^n.

$r = 1$: perfect +ve corr^n.

* Simple linear Regression $[y = f(x)]$:-

Normal Equations :-

$y = w_0 + w_1 x.$

$\Sigma y = w_0 n + w_1 \Sigma x$

$\Sigma xy = w_0 \Sigma x + w_1 \Sigma x^2$

Sum of squared errors :- [gives predictive model to describe rel^n. bet^n. 2 variables]

$$SSE = \Sigma (y_i - \hat{y}_i)^2$$

$$= \Sigma [y_i - (w_0 + w_1 x_i)]^2$$

Matrix form :-

$$\begin{bmatrix} \Sigma y \\ \Sigma xy \end{bmatrix} = \begin{bmatrix} n & \Sigma x \\ \Sigma x & \Sigma x^2 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \end{bmatrix}$$

$$\underset{(1\times 3)}{B} = \underset{(3\times 3)}{A}\ \underset{(3\times 1)}{X}$$

* Multiple Linear Regression :- $y = f(x_1, x_2, x_3, \dots)$

⇒ with Two independent variables :-

$$y = f(x_1, x_2)$$
$$y = w_0 + w_1 x_1 + w_2 x_2$$

$$SSE : \Sigma (y - \hat{y})^2$$
$$= \Sigma [y - (w_0 + w_1 x_1 + w_2 x_2)]^2$$

$$\Sigma y = w_0 n + w_1 \Sigma x_1 + w_2 \Sigma x_2$$

$$\Sigma x_1 y = w_0 \Sigma x_1 + w_1 \Sigma x_1^2 + w_2 \Sigma x_1 x_2$$

$$\Sigma x_2 y = w_0 \Sigma x_2 + w_1 \Sigma x_1 x_2 + w_2 \Sigma x_2^2$$

⇒ with more than two independent variables :-

$$y = f(x_1, x_2, \dots x_n)$$
$$SSE : \Sigma (y - \hat{y})^2$$

$$y = w_0 + w_1 x_1 + w_2 x_2 + \dots + w_n x_n.$$

* Accuracy of Regression model :- [R^2 & Adjusted R^2 value]

$$R^2 = 1 - \frac{RSS}{TSS}$$

$$RSS \rightarrow \sum_i (y_i - \underbrace{(w_0 + w_1 x_i)}_{\hat{y}})^2$$

$$TSS \rightarrow \sum_i (y_i - \bar{y})^2$$

^mean of y

RSS → Sum of squared errors due to regression

TSS → Sum of squared errors total.

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* **Adjusted R^2 value:**

$$\text{Adjusted } R^2 \text{ value} = 1 - \frac{(1-R^2)(N-1)}{(N-P-1)}$$

$N \rightarrow$ total sample size
$P \rightarrow$ No. of independent variable
$R^2 \rightarrow R^2$ value

* **Non-Linear regression:**

(I) $xy = w_0 + w_1 x$

$$y = \frac{w_0}{x} + w_1$$

$$1/x = X$$

$$y = X w_0 + w_1 \quad [SLR]$$

$$\left.\begin{aligned} \Sigma y &= w_1 n + w_0 \Sigma X \\ \Sigma X y &= w_1 \Sigma X + w_0 \Sigma X^2 \end{aligned}\right\} X = \frac{1}{x}$$

(II) $y = ae^{bx}$

$$\underbrace{\log y}_{Y} = \underbrace{\log a}_{A} + b \underbrace{\log x}_{X}$$

$Y = A + bX \rightarrow$ linear

$$\Sigma Y = An + b \Sigma X$$

$$\Sigma XY = A \Sigma X + b \Sigma X^2$$

* Polynomial Regression

$$y = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$$

$$y = \omega_1 x + \omega_2 x^2$$

$$\sum xy = \omega_1 \sum x^2 + \omega_2 \sum x^3$$
$$\sum x^2 y = \omega_1 \sum x^3 + \omega_2 \sum x^4 .$$

$$SSE = \sum (y - \hat{y})^2$$
$$= \sum y - (\omega_1 x + \omega_2 x^2)$$

* Testing of Hypothesis: Summary :-

Testing of hypothesis

- mean
  - one mean
    - Large Sample: $Z = \dfrac{\bar{x} - \mu}{\sigma / \sqrt{n}}$
    - small sample: $t = \dfrac{\bar{x} - \mu}{s / \sqrt{n}}$, $\nu = n - 1$ (dof)
  - Two mean
    - Large sample: $Z = \dfrac{\bar{x}_1 - \bar{x}_2}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}}$
    - small sample: $t = \dfrac{\bar{x}_1 - \bar{x}_2}{\sqrt{\dfrac{S_1^2}{n_1} + \dfrac{S_2^2}{n_2}}}$
- proportion
  - one: $Z = \dfrac{p - P}{\sqrt{\dfrac{PQ}{n}}}$
  - Two: $Z = \dfrac{\left(\dfrac{x_1}{n_1}\right) - \left(\dfrac{x_2}{n_2}\right)}{\sqrt{\hat{p}(1-\hat{p})\left(\dfrac{1}{n_1} + \dfrac{1}{n_2}\right)}}$, $\hat{p} = \dfrac{x_1 + x_2}{n_1 + n_2}$
  - Several: $\chi^2 = \sum \dfrac{(O-E)^2}{E}$